Topic 4 Vectors

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4.1 Introduction to Vectors

4.1 Introduction to Vectors: Properties of Vectors

- Recall, from our work on numbers and sets, the set

$$\mathbb{R}^2 = \mathbb{R} \times \mathbb{R} = \{(a_1, a_2) \mid a_1 \in \mathbb{R} \text{ and } a_2 \in \mathbb{R}\}.$$

- More generally, for any natural number n we consider the set

$$\mathbb{R}^n = \{(a_1, a_2, \ldots, a_n) \mid a_1, a_2, \ldots, a_n \in \mathbb{R}\},$$

  usually called n-dimensional space.
- A <u>vector</u> is an element of $\mathbb{R}^n$.
- For example:
  - $(3, 1, 0)$ and $(-1, -1, -1)$ are vectors in $\mathbb{R}^3$;
  - $(0, 0)$ and $(1, 5)$ are vectors in $\mathbb{R}^2$;
  - $(0, 1, 2, 3, 4, 5, 6, 7, 8, 9)$ is a vector in $\mathbb{R}^{10}$.

4.1 Introduction to Vectors: Vector Arithmetic

- In our work in complex numbers, we defined various operations for complex numbers
- We later saw that these operations made sense *geometrically*
- We do the same for vectors.

Definition 4.1

Let $\mathbf{a} = (a_1, a_2, \ldots, a_n)$ and $\mathbf{b} = (b_1, b_2, \ldots, b_n)$ be vectors in $\mathbb{R}^n$. Let $\lambda \in \mathbb{R}$. We define the following operations

- <u>addition</u>

$$\begin{aligned}\mathbf{a} + \mathbf{b} &= (a_1, a_2, \ldots, a_n) + (b_1, b_2, \ldots, b_n) \\ &= (a_1 + b_1, a_2 + b_2, \ldots, a_n + b_n)\end{aligned}$$

- <u>subtraction</u>

$$\begin{aligned}\mathbf{a} - \mathbf{b} &= (a_1, a_2, \ldots, a_n) - (b_1, b_2, \ldots, b_n) \\ &= (a_1 - b_1, a_2 - b_2, \ldots, a_n - b_n)\end{aligned}$$

- <u>scalar multiplication</u>

$$\lambda\mathbf{a} = \lambda(a_1, a_2, \ldots, a_n) = (\lambda a_1, \lambda a_2, \ldots, \lambda a_n)$$

Example 4.2

Let $\mathbf{v} = (3, -2)$ and $\mathbf{w} = (1, 5)$. Compute

(a) $\mathbf{v} + \mathbf{w}$

(b) $\mathbf{v} - \mathbf{w}$

(c) $\mathbf{w} - \mathbf{v}$

(d) $3\mathbf{v}$

4.1 Introduction to Vectors: Vector Arithmetic

Homework 32

Let $\mathbf{u} = (1, -3, 2)$ *and* $\mathbf{v} = (-1, 0, -4)$*. Repeat (a)–(d) from the previous page.*

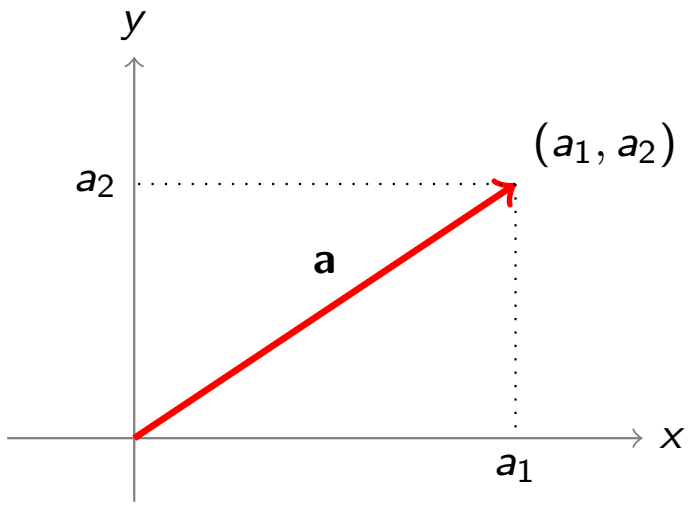
Note: One may only perform arithmetic with pairs of vectors of the same dimension. The following does not have any mathematical meaning:

$$(1, 2, 3, 4) + (5, 6) = ???$$

4.1 Geometry of Vectors: Vectors as Arrows

- For vectors in $\mathbb{R}^2$ and $\mathbb{R}^3$, we can visualise vectors as directed line segments or <u>arrows</u>.

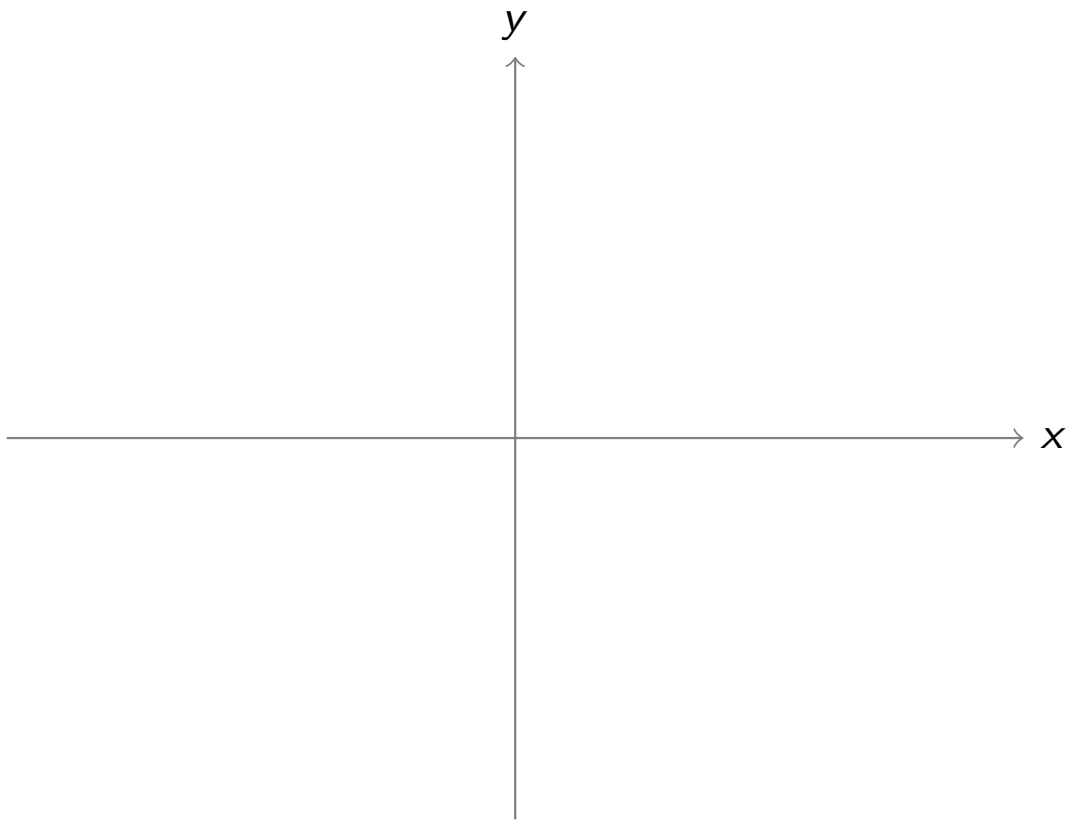
- $(a_1, a_2) \in \mathbb{R}^2$ is represented by the arrow from the origin $(0, 0)$ to the point (a_1, a_2).



4.1 Introduction to Vectors: Vectors as Arrows

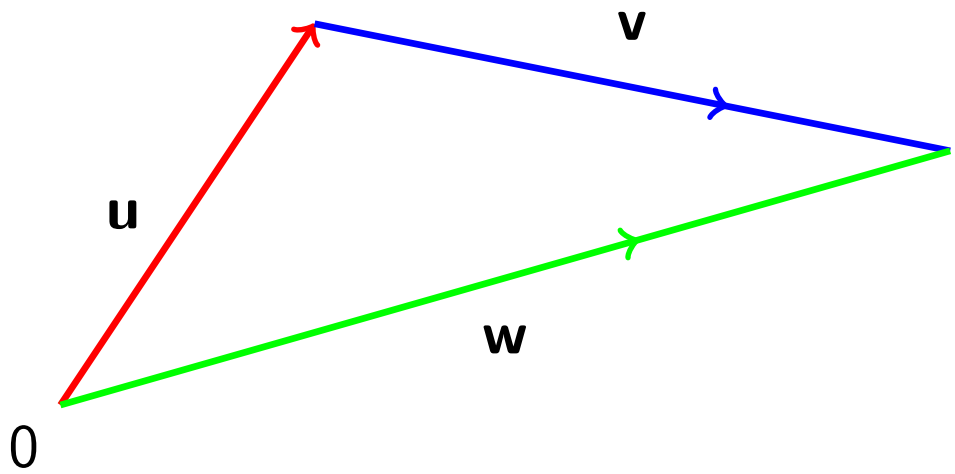
- With this interpretation we add vectors by placing the tail of the second vector at the tip of the first vector.

Let $\mathbf{u} = (3, -2)$ and $\mathbf{v} = (1, 5)$.



4.1 Introduction to Vectors: Vectors as Arrows

- For vectors $\mathbf{u}$ and $\mathbf{v}$ the vector $\mathbf{w} = \mathbf{u} + \mathbf{v}$ is represented by the arrow starting at the base of $\mathbf{u}$ and ending at the tip of $\mathbf{v}$.



4.1 Introduction to Vectors: Vectors as Arrows

- Notice that in our picture, the vector **v** does not start at the origin.
- Given points $A = (x_1, x_2)$ and $B = (y_1, y_2)$ in $\mathbb{R}^2$, the vector $\overrightarrow{AB}$ is defined by

$$\begin{aligned}\overrightarrow{AB} &= (y_1, y_2) - (x_1, x_2) \\ &= (y_1 - x_1, y_2 - x_2)\end{aligned}$$

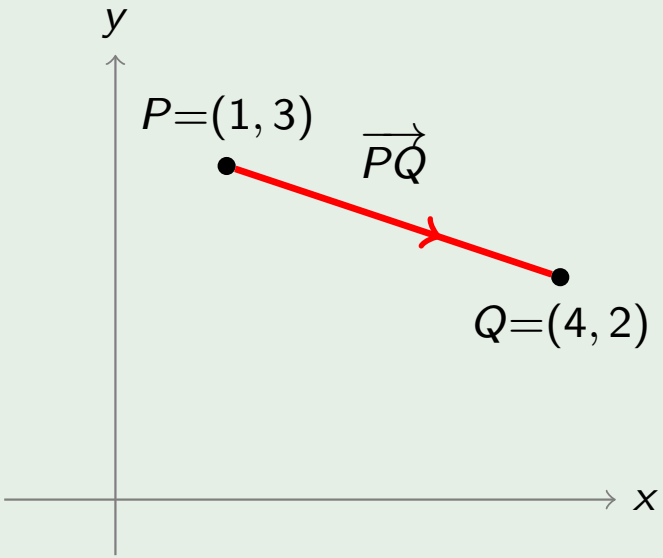
- In this notation, $P \in \mathbb{R}^2$ may be written as $\overrightarrow{OP}$, where $O = (0, 0)$ denotes the origin.
- $\overrightarrow{OP}$ is called the <u>position vector</u> of P.

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}.$$

4.1 Introduction to Vectors: Vectors as Arrows

Example 4.3

Express the directed line segment $\overrightarrow{PQ}$ shown in the picture as a vector by subtracting position vectors.

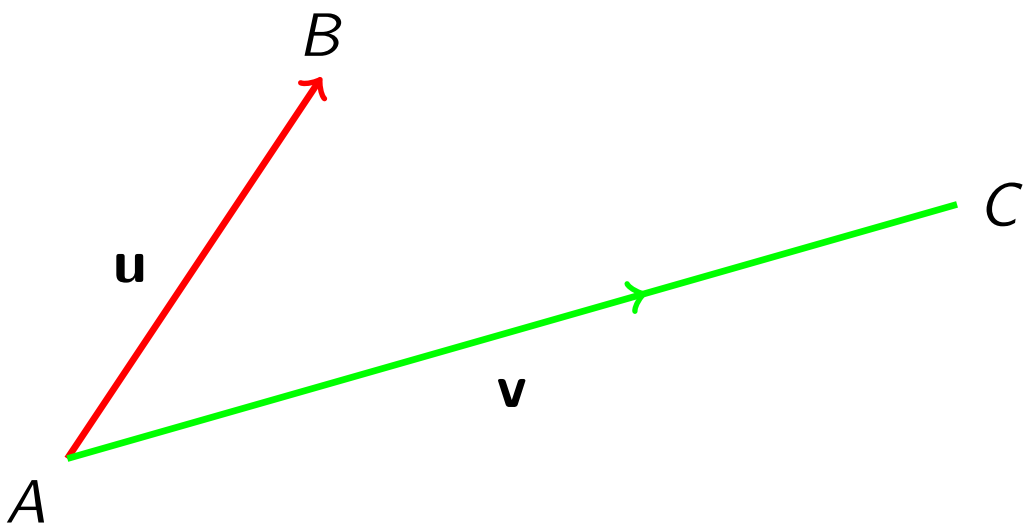


4.1 Introduction to Vectors: Geometric Subtraction

- Once we know how to add vectors geometrically, we can subtract geometrically using a similar diagram.

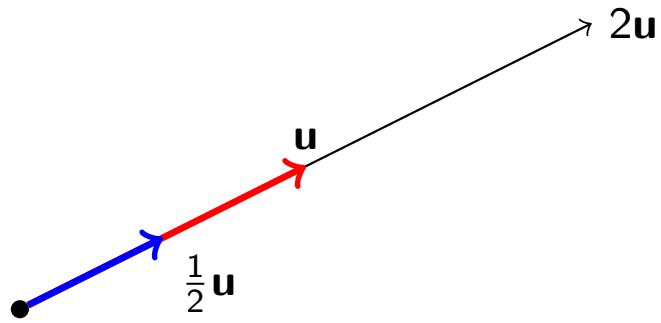
Example 4.4

Illustrate the vector $\mathbf{v} - \mathbf{u}$ in the diagram.

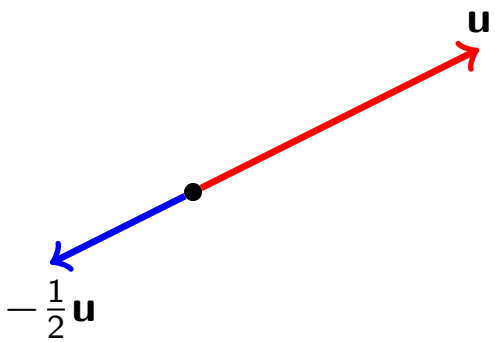


4.1 Introduction to Vectors: Geometric Scalar Multiplication

- Multiplying by $\lambda > 0$ stretches or shrinks $\mathbf{u}$.



- Multiplying by $\lambda < 0$ stretches or shrinks $\mathbf{u}$ and reverses the direction of $\mathbf{u}$

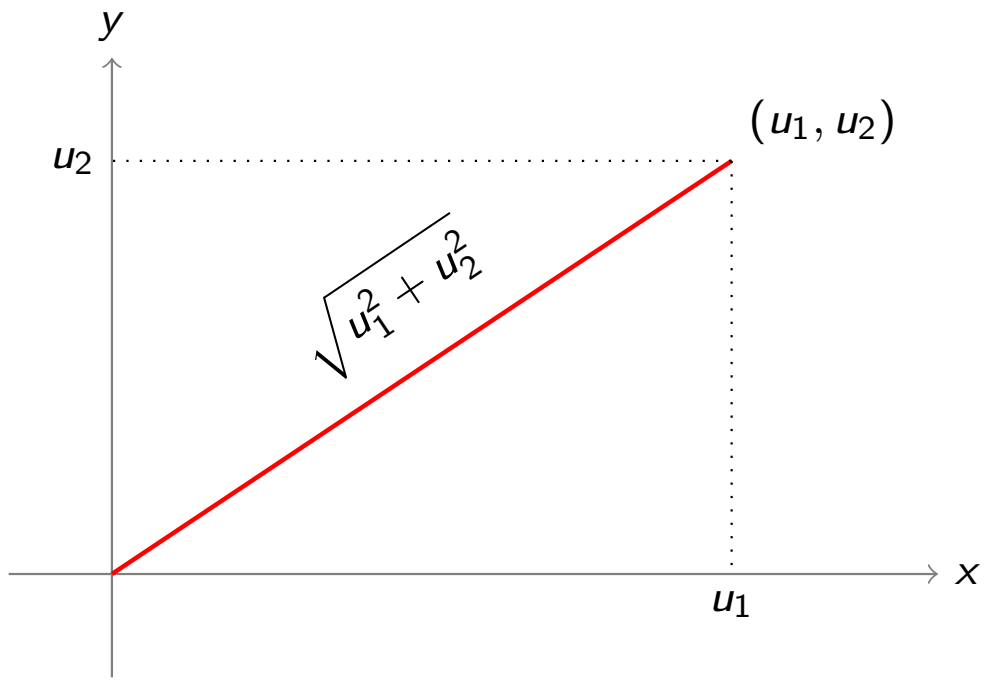


Definition 4.5

Let $\mathbf{u}$ and $\mathbf{v}$ be non-zero vectors. We say $\mathbf{u}$ and $\mathbf{v}$ are <u>parallel</u> when there exists $\lambda \in \mathbb{R}$ such that $\mathbf{u} = \lambda\mathbf{v}$.

4.1 Introduction to Vectors: Vector Length

- Intuitively, the length of a vector is the length of the arrow that represents the vector graphically.



4.1 Introduction to Vectors: Vector Length

Definition 4.6

Let $\mathbf{u} \in \mathbb{R}^2$ the <u>norm</u> of $\mathbf{u}$ is denoted $\|\mathbf{u}\|$, is given by

$$\|\mathbf{u}\| = \sqrt{u_1^2 + u_2^2}$$

- We can extend this definition to a vector $\mathbf{u}$ in $\mathbb{R}^n$:

$$\|\mathbf{u}\| = \sqrt{u_1^2 + u_2^2 + \cdots + u_n^2}$$

4.1 Introduction to Vectors: Vector Length

Example 4.7

Find the length of the vectors (a) $(1, 2)$ (b) $(-1, 3, -2)$.

Homework 33

Find the length of the vectors (a) $(-2, -1)$ *(b)* $(-3, 6, -2)$.

4.1 Introduction to Vectors: Vector Length and Distance

This definition of length gives us exactly the properties we expect:

1. Length is never negative. Only the 0 vector has length 0.
2. The length of a scalar multiple is the absolute value of the scalar multiple times the length of the vector
3. Length obeys the triangle inequality.

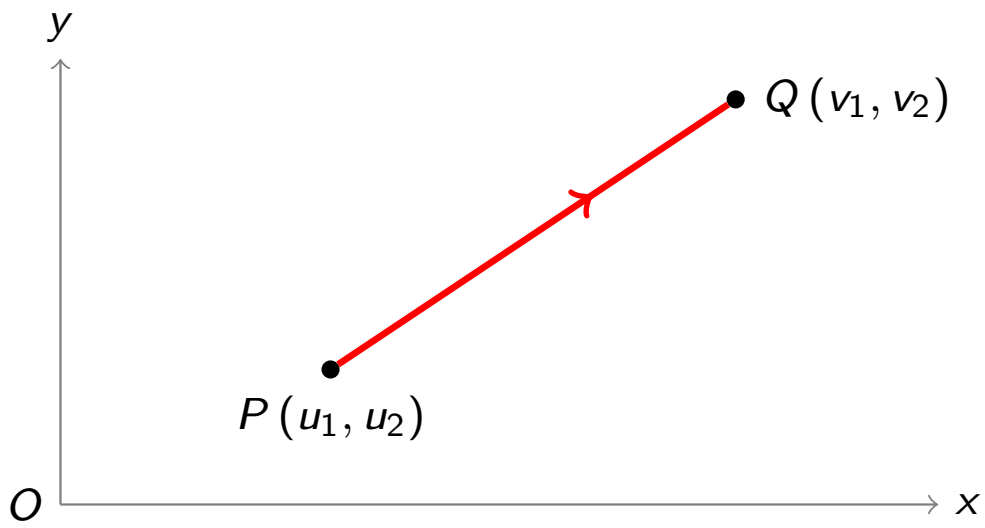
Theorem 4.8 (Properties of length)

For any $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ *and* $\lambda \in \mathbb{R}$ *:*

1. $\|\mathbf{u}\| \geq 0$ *and* $\|\mathbf{u}\| = 0$ *exactly when* $\mathbf{u} = \mathbf{0}$.
2. $\|\lambda \mathbf{u}\| = |\lambda| \|\mathbf{u}\|$.
3. $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$.

4.1 Introduction to Vectors: Vector Length and Distance

- Similar to complex numbers $\|\mathbf{v} - \mathbf{u}\|$ gives the distance between (the tips of) $\mathbf{v}$ and $\mathbf{u}$.
- If $P = (u_1, u_2)$ and $Q = (v_1, v_2)$ are points in $\mathbb{R}^2$, the distance between them is the length of the vector $\overrightarrow{PQ}$:



- We know that $\overrightarrow{PQ} = (v_1 - u_1, v_2 - u_2)$, so the distance between P and Q is

$$\left\|\overrightarrow{PQ}\right\| = \sqrt{(v_1 - u_1)^2 + (v_2 - u_2)^2}.$$

4.1 Introduction to Vectors: Vector Length and Distance

- We can extend this idea to $\mathbb{R}^3$ in the way we expect.

Example 4.9

Find the distance between the points $P = (-2, 1, 0)$ and $Q = (3, -1, 1)$.

4.1 Introduction to Vectors: Vector Length and Distance

Homework 34

Find the distance between the points $A = (-3, 1, -1)$ *and* $B = (2, 0, -1)$.

4.1 Introduction to Vectors: Unit Vectors

- So far we have built up operations (and geometric interpretations) in $\mathbb{R}^n$ that replicate similar ideas in $\mathbb{R}$.
- In $\mathbb{R}$ every real number can be written as a scalar multiple of 1.
- When we look at vectors with length 1 in $\mathbb{R}^n$ we see the same behaviour.

4.1 Introduction to Vectors: Unit Vectors

- Any vector of length 1 is called a <u>unit vector</u>.
- We denote this unit vector in the direction of $\mathbf{u}$ by $\hat{\mathbf{u}}$.

- Every vector $\mathbf{u}$ can be expressed as $\mathbf{u} = \lambda\hat{\mathbf{u}}$ with $\lambda = \|\mathbf{u}\|$.

4.1 Introduction to Vectors: Unit Vectors

Definition 4.10

Let $\mathbf{u}$ be a non-zero vector in $\mathbb{R}^n$. The <u>unit vector</u> of $\mathbf{u}$ is the vector of length 1 that is in the direction of $\mathbf{u}$. The unit vector of $\mathbf{u}$ is denoted as $\hat{\mathbf{u}}$.

Theorem 4.11

If $\mathbf{u}$ *is a non-zero vector in* $\mathbb{R}^n$*, then*

$$\hat{\mathbf{u}} = \frac{1}{\|\mathbf{u}\|}\mathbf{u}$$

is the unit vector of $\mathbf{u}$*.*

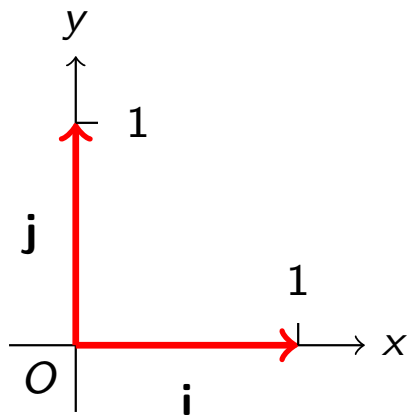
4.1 Introduction to Vectors: Unit Vectors

Example 4.12

Let $\mathbf{v}$ be the vector from $A = (2, 0, -1)$ to $B = (1, 2, -3)$.
Find two unit vectors that are parallel to $\mathbf{v}$.

4.1 Introduction to Vectors: Standard Unit Vectors

- The unit vectors $\mathbf{i} = (1, 0)$ and $\mathbf{j} = (0, 1)$ are called the <u>standard unit vectors in $\mathbb{R}^2$</u>.
- Every element of $\mathbb{R}^2$ can be written as a linear combination of $\mathbf{i}$ and $\mathbf{j}$.

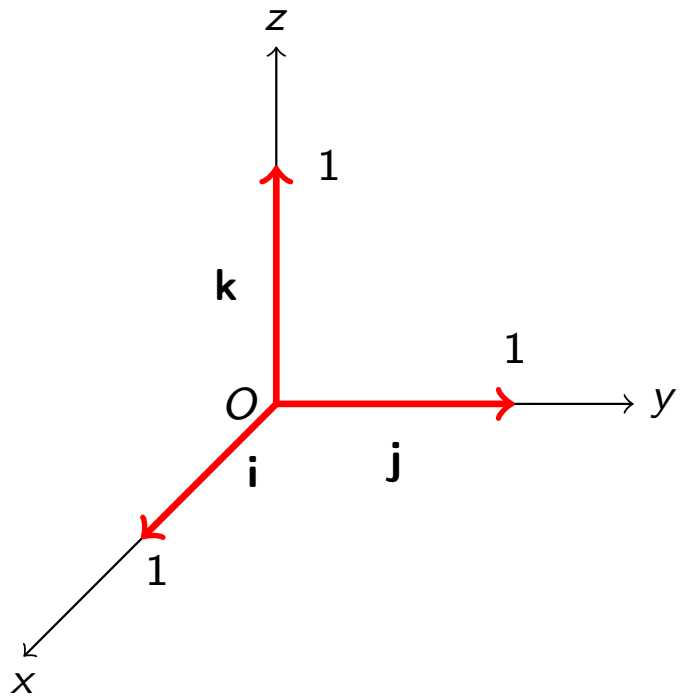


Example 4.13

Express the vectors $\mathbf{v} = (1, 2)$ and $\mathbf{u} = (-1, 3)$ as a linear combination of $\mathbf{i}$ and $\mathbf{j}$

4.1 Introduction to Vectors: Standard Unit Vectors

- In $\mathbb{R}^3$, the standard unit vectors are $\mathbf{i} = (1, 0, 0)$, $\mathbf{j} = (0, 1, 0)$ and $\mathbf{k} = (0, 0, 1)$.



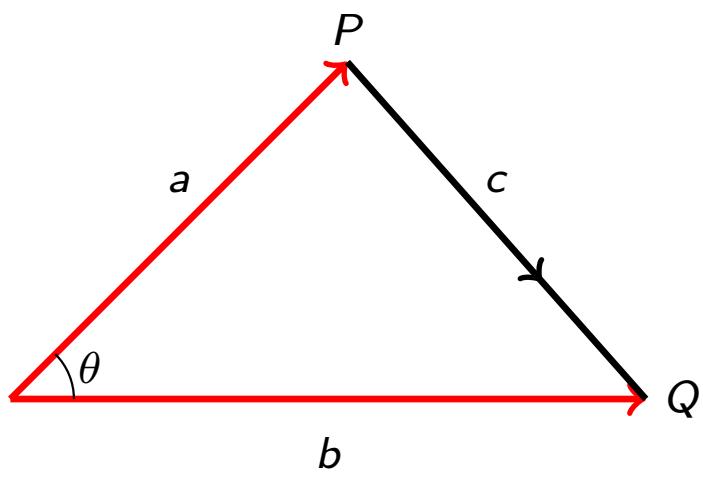
Homework 35

Express $(-3, 1, 5) \in \mathbb{R}^3$ *as a linear combination of* $\mathbf{i}$, $\mathbf{j}$ *and* $\mathbf{k}$.

4.2 The Scalar Product

4.2 The Scalar Product: Law of Cosines

- One tool that we can use to examine the angle between vectors is the <u>law of cosines</u>

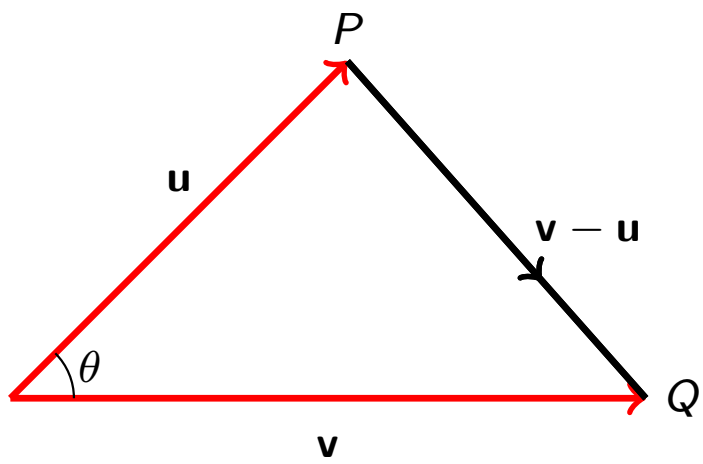


- The Law of Cosines:

$$c^2 = a^2 + b^2 - 2ab\cos\theta$$

4.2 The Scalar Product: Law of Cosines

- One tool that we can use to examine the angle between vectors is the <u>law of cosines</u>



- The Law of Cosines:

$$\|\mathbf{v}-\mathbf{u}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\|\mathbf{u}\|\|\mathbf{v}\|\cos(\theta).$$

4.2 The Scalar Product

Definition 4.14

Let $\mathbf{u}$ and $\mathbf{v}$ be vectors in $\mathbb{R}^n$: with $\mathbf{u} = (u_1, u_2, \ldots, u_n)$ and $\mathbf{v} = (v_1, v_2, \ldots, v_n)$. The <u>scalar product</u> of $\mathbf{u}$ and $\mathbf{v}$ is defined as:

$$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2 + \ldots + u_n v_n.$$

- For instance, in $\mathbb{R}^3$ we have $\mathbf{u} = (u_1, u_2, u_3)$, $\mathbf{v} = (v_1, v_2, v_3)$ and

$$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2 + u_3 v_3.$$

4.2 The Scalar Product: Examples in $\mathbb{R}^3$

Example 4.15

Let $\mathbf{u} = (2, 3, -1)$ and $\mathbf{v} = (4, 5, 0)$. Calculate $\mathbf{u} \cdot \mathbf{v}$.

4.2 The Scalar Product: Examples in $\mathbb{R}^3$

Homework 36

Calculate $\mathbf{u} \cdot \mathbf{v}$ *for* $\mathbf{u} = (3, 2, 1)$ *and* $\mathbf{v} = (1, 1, 3)$

4.2 The Scalar Product: Properties

Theorem 4.16

Let $\mathbf{u}$, $\mathbf{v}$ *and* $\mathbf{w}$ *be vectors in* $\mathbb{R}^n$ *and let* $\lambda \in \mathbb{R}$.

1. *The scalar product is commutative; that is*

$$\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}.$$

2. *The scalar product is (left and right) distributive; that is*

$$\mathbf{v} \cdot (\mathbf{u} + \mathbf{w}) = \mathbf{v} \cdot \mathbf{u} + \mathbf{v} \cdot \mathbf{w} \quad \text{and} \quad (\mathbf{u} + \mathbf{w}) \cdot \mathbf{v} = \mathbf{u} \cdot \mathbf{v} + \mathbf{w} \cdot \mathbf{v}.$$

3. $\lambda(\mathbf{u} \cdot \mathbf{v}) = (\lambda \mathbf{u}) \cdot \mathbf{v} = \mathbf{u} \cdot (\lambda \mathbf{v})$.
4. *The scalar product of a vector with itself yields* $\mathbf{u} \cdot \mathbf{u} = \|\mathbf{u}\|^2$.

4.2 The Scalar Product: Angles

Theorem 4.17

If θ is the angle in $[0, \pi]$ between non-zero vectors $\mathbf{u}$ and $\mathbf{v}$, then:

$$\cos(\theta) = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}$$

and

$$\theta = \arccos\left(\frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}\right).$$

4.2 The Scalar Product: Angles

Example 4.18

Find the angle θ between the vectors $(-2, 1, 2)$ and $(1, -1, 0)$.

4.2 The Scalar Product: Angles

Example 4.19

Find the cosine of the angle θ between $\mathbf{u} = (1, -2, 0)$ and $\mathbf{v} = (3, 1, -2)$.

4.2 The Scalar Product: Angles

Homework 37

Find the angle between $\mathbf{u} = (-3, 0, 1)$ *and* $\mathbf{v} = (2, -2, 1)$.

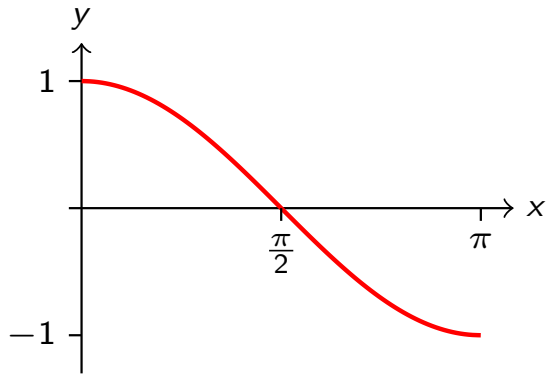
4.2 The Scalar Product: Angles

- We have seen that if θ is the angle in $[0, \pi]$ between non-zero vectors $\mathbf{u}$ and $\mathbf{v}$, then:

$$\cos(\theta) = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|}$$

  and $\|\mathbf{u}\|\|\mathbf{v}\| > 0$.

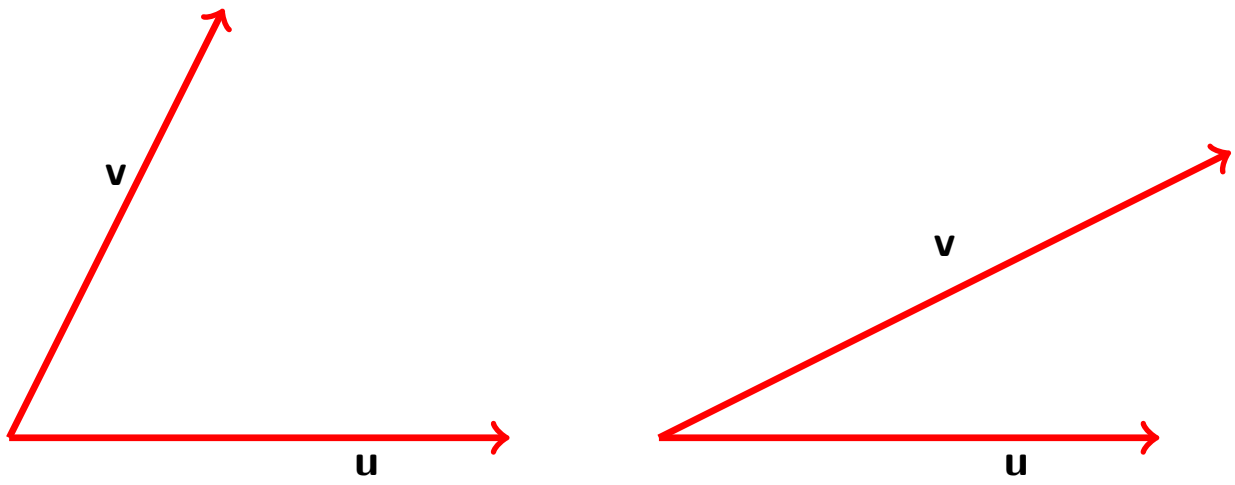
- The graph of $\cos(\theta)$ restricted to $[0, \pi]$ therefore tells us that:
  - θ is acute precisely when $\mathbf{u} \cdot \mathbf{v} > 0$.
  - θ is obtuse precisely when $\mathbf{u} \cdot \mathbf{v} < 0$.
  - θ is $\frac{\pi}{2}$ precisely when $\mathbf{u} \cdot \mathbf{v} = 0$.



4.3 Vector Projections: An Application of the Scalar Product

4.3 Vector Projections

- Idea: For any pair of non-zero vectors $\mathbf{u}$ and $\mathbf{v}$ we can express $\mathbf{v}$ as a sum $\mathbf{v} = k\mathbf{u} + \mathbf{w}$, where $\mathbf{w}$ is perpendicular to $\mathbf{u}$

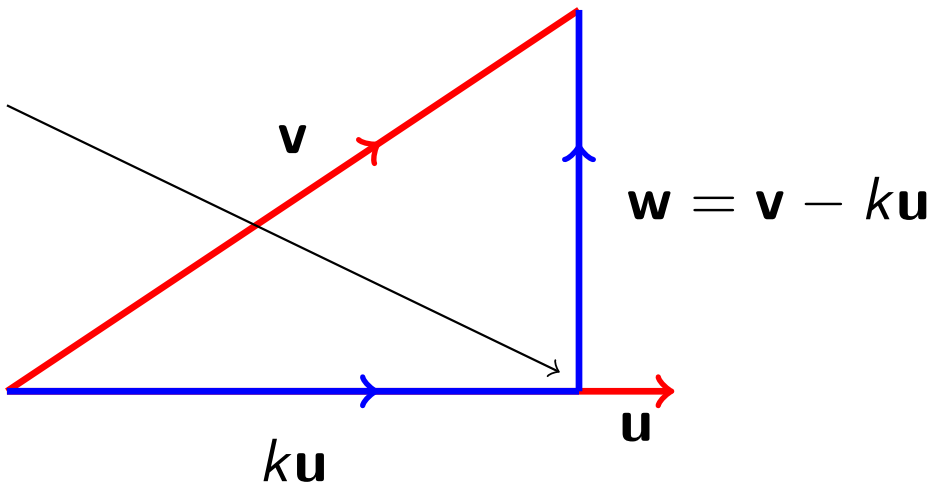


- we can express $\mathbf{v}$ as $k\mathbf{u} + \mathbf{w}$ by letting $\mathbf{w} = \mathbf{v} - k\mathbf{u}$.

4.3 Vector Projections: Sums of Vectors

- We want to write $\mathbf{v}$ as the sum of a vector $\mathbf{v}_{\parallel}$ parallel to $\mathbf{u}$ and another vector $\mathbf{v}_{\perp}$ perpendicular to $\mathbf{u}$.

Choose k to make angle $\frac{\pi}{2}$.



4.3 Vector Projections: Sums of Vectors

- Recall: vectors are perpendicular precisely when their scalar product is 0.
- Thus we want to find k such that

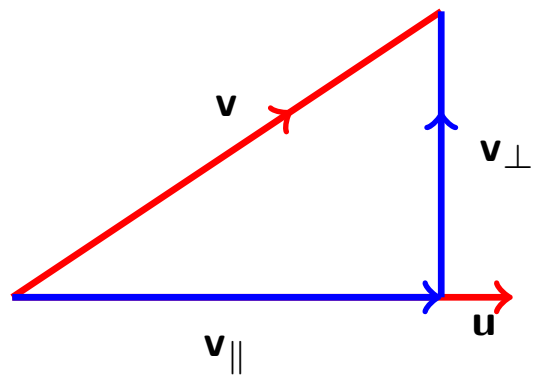
$$\mathbf{u} \cdot (\mathbf{v} - k\mathbf{u}) = 0.$$

- The parallel and perpendicular projections of $\mathbf{v}$ onto $\mathbf{u}$ are given by

$$\mathbf{v}_{\parallel} = k\mathbf{u},$$
$$\mathbf{v}_{\perp} = \mathbf{v} - k\mathbf{u}$$

where k is the solution of

$$\mathbf{u} \cdot (\mathbf{v} - k\mathbf{u}) = 0.$$



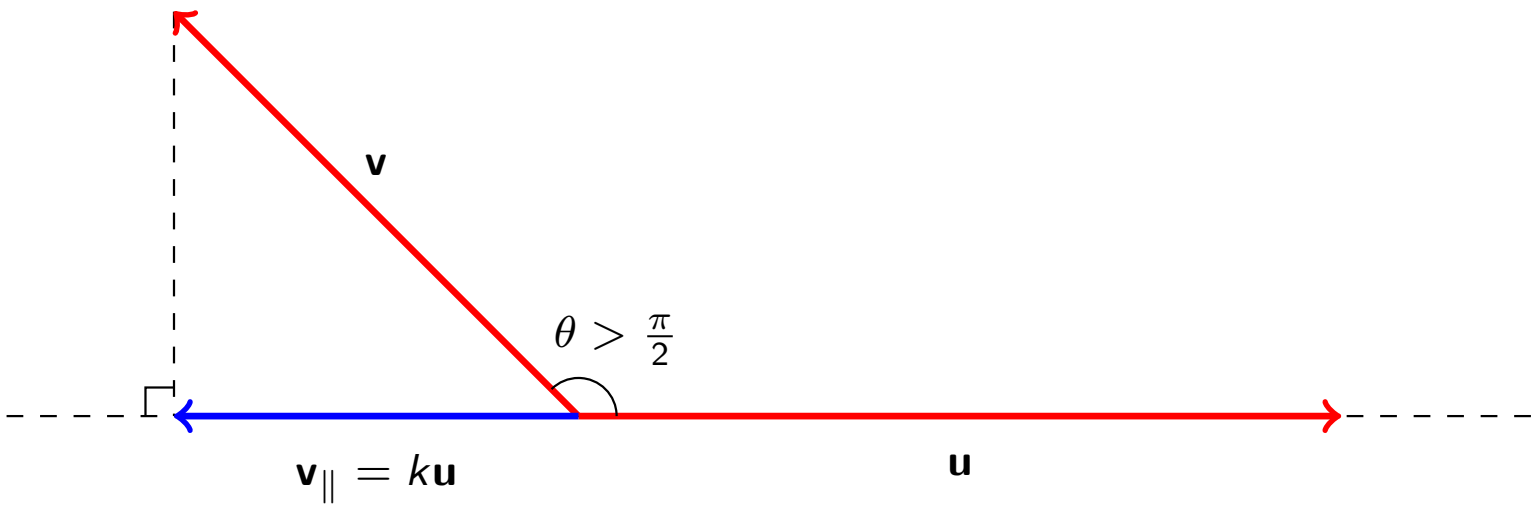
4.3 Vector Projections: Sums of Vectors

Example 4.20

Let $\mathbf{u} = (3, 1, -2)$ and $\mathbf{v} = (1, 0, 5)$. Find the parallel and perpendicular projections of $\mathbf{v}$ onto $\mathbf{u}$.

4.3 Vector Projections: Reversing the Process

- In the previous example, we obtained a negative k value.
- This means that the vector $k\mathbf{u}$ points in the opposite direction to $\mathbf{u}$, so the angle between $\mathbf{u}$ and $\mathbf{v}$ is obtuse.



- When k is positive, the angle θ is acute.

4.3 Vector Projections

Homework 38

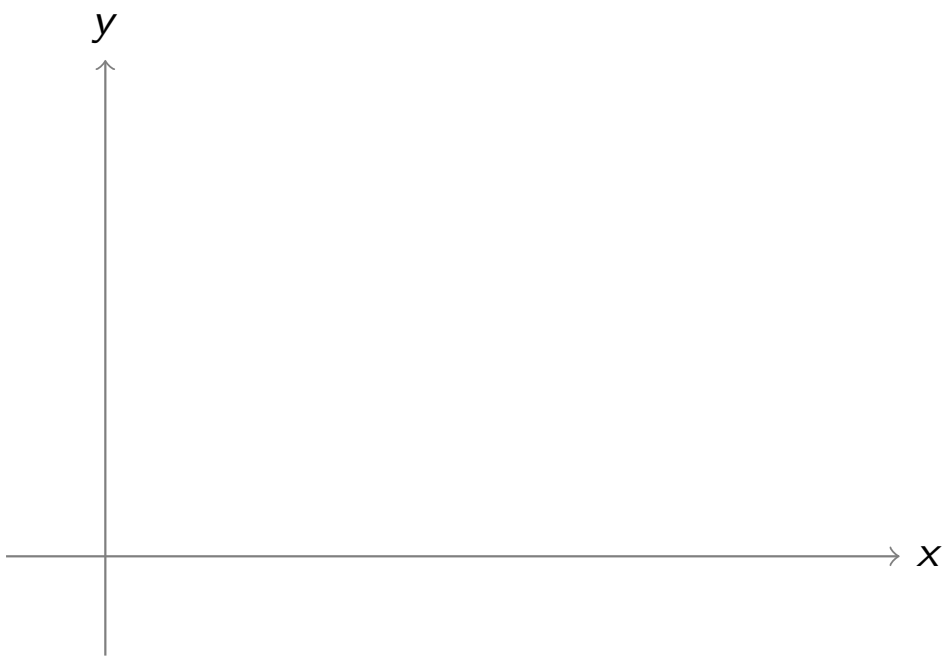
Let $\mathbf{u} = (1, 1, 0)$ *and* $\mathbf{v} = (1, -2, 3)$*. Find the parallel and perpendicular vector projections of* $\mathbf{v}$ *onto* $\mathbf{u}$*. Draw them on a single diagram with* $\mathbf{u}$ *and* $\mathbf{v}$

4.3 Vector Projections: Points and Lines

- We can use vector projections to find the closest point on a line L to a given point P.

Example 4.21

Let $P = (1, 3)$ and let L be the line passing through $A = (0, 1)$ and $B = (4, 3)$. Find the closest point Q on the line L to P and hence find the distance from P to L.



4.4 Parametric Curves

4.4 Parametric Curves

Just as we can think about the distinction between constants (4, 11, π) and functions ($f(x) = 2x + \pi$), we can also think about the distinction between vectors and functions whose outputs are vectors.

$$\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j}, \qquad \text{where } t \in \mathbb{R}.$$

Here $\mathbf{r}$ is a function $\mathbf{r}\colon \mathbb{R} \longrightarrow \mathbb{R}^2$, since for each $t \in \mathbb{R}$, $\mathbf{r}(t)$ is a vector in $\mathbb{R}^2$. This is called a <u>vector-valued</u> function of a real variable (which we often think of as time).

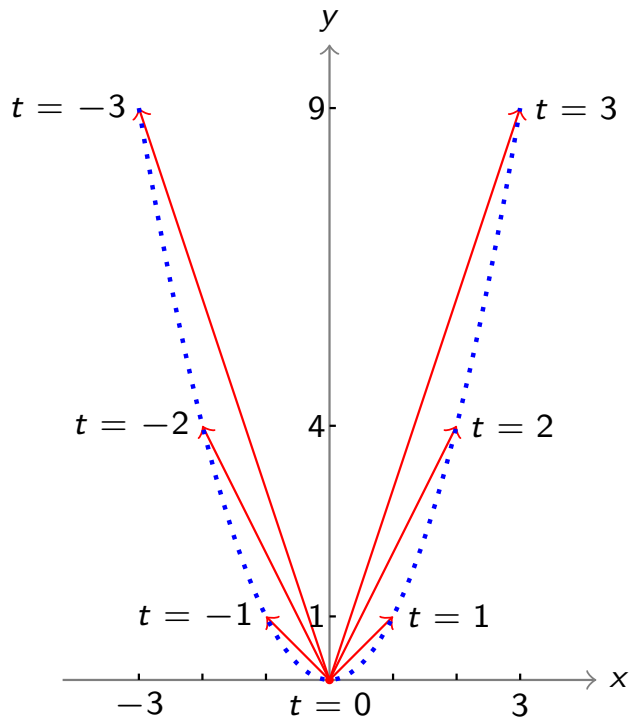
The functions $x(t)$ and $y(t)$ are called <u>parametric functions</u>, since they depend on the parameter t. The resulting curve that $\mathbf{r}(t)$ traces out in $\mathbb{R}^2$ is called an <u>parametric curve</u>.

4.4 Parametric Curves

- Consider the formula $\mathbf{r}(t) = t\mathbf{i} + t^2\mathbf{j}$, for $t \in \mathbb{R}$. In this example $x(t) = t$ and $y(t) = t^2$.
- To sketch the curve that $\mathbf{r}(t)$ traces out, we need to know the values of x and y for different values of t.

- For each point t we get a vector $\mathbf{r}(t)$.

4.4 Parametric Curves



- The path of **r** is the curve traced out by the heads of the vectors as t varies through dom(**r**)
- In the diagram the path is indicated by the dotted line.

4.4 Parametric Curves: Equation of a Path

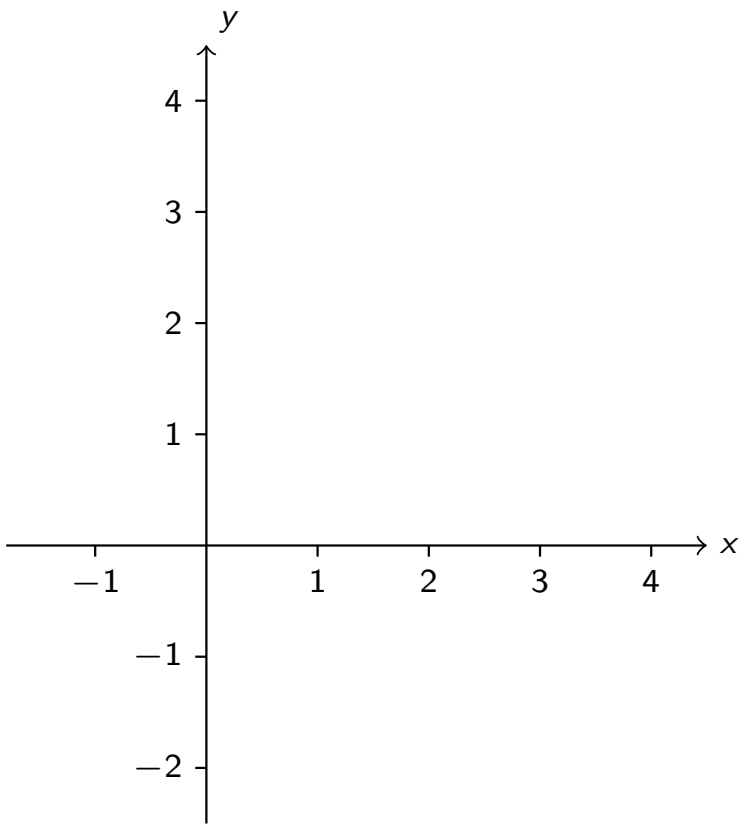
Example 4.22

Find the Cartesian equation of the path of a particle whose position is given by $\mathbf{r} : \mathbb{R} \longrightarrow \mathbb{R}^2$ with formula

$$\mathbf{r}(t) = (t^2 - t)\mathbf{i} + 3t\,\mathbf{j}.$$

Sketch the path, indicating the direction of increasing t.

4.4 Parametric Curves: Equation of a Path



4.4 Parametric Curves: Equation of a Path

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Find the Cartesian equation of the path of a particle whose position is given by $\mathbf{r}(t) = t\mathbf{i} + (t^2 + 1)\mathbf{j}$, *for* $t \geq 0$.

4.4 Parametric Curves: Equation of a Path

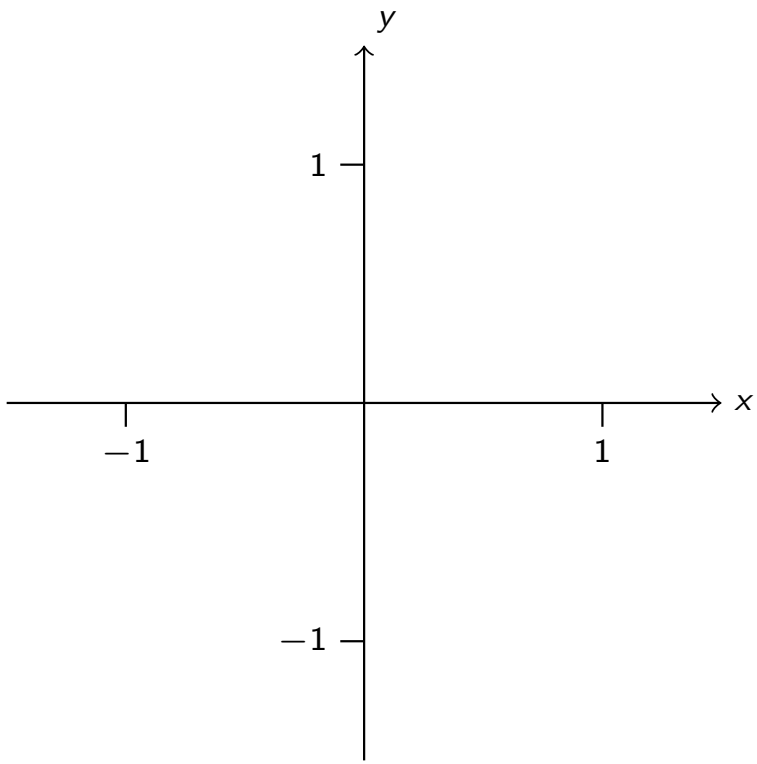
Example 4.23

Find the equation of the path of a particle whose position is given by

$$\mathbf{r}(t) = \cos(t)\mathbf{i} + \sin(t)\mathbf{j}, \qquad \text{for } t \in \mathbb{R}.$$

Sketch the path.

4.4 Parametric Curves: Equation of a Path



4.4 Parametric Curves: Cartesian Equations and Parametric Curves

- In Example 4.23, the path of **r** was a circle, which we could have sketched from the Cartesian equation $x^2 + y^2 = 1$.
- But the parametric curve tells us much more about the motion of the particle. E.g.,:

- The parametric curve tells us how the particle is moving.

4.4 Parametric Curves

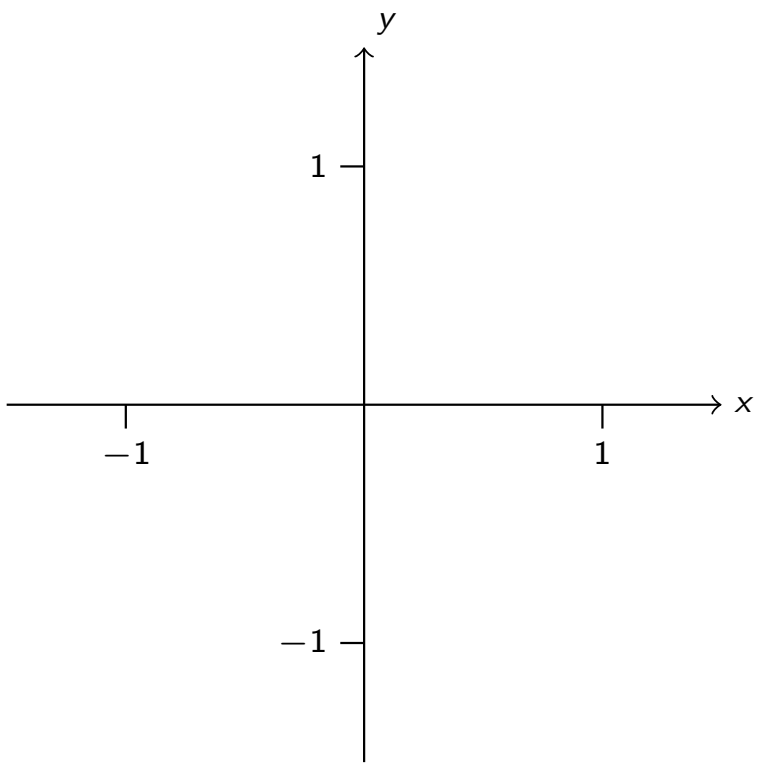
Example 4.24

Find the equation of the path of a particle whose position is given by

$$\mathbf{r}(t) = \sin(t)\mathbf{i} + \cos(t)\mathbf{j}, \qquad \text{for } t \in \mathbb{R}.$$

Sketch the path.

4.4 Parametric Curves



- Note that where the motion in Example 4.23 was anticlockwise, the motion in this example is clockwise.

4.4 Parametric Curves

Example 4.25

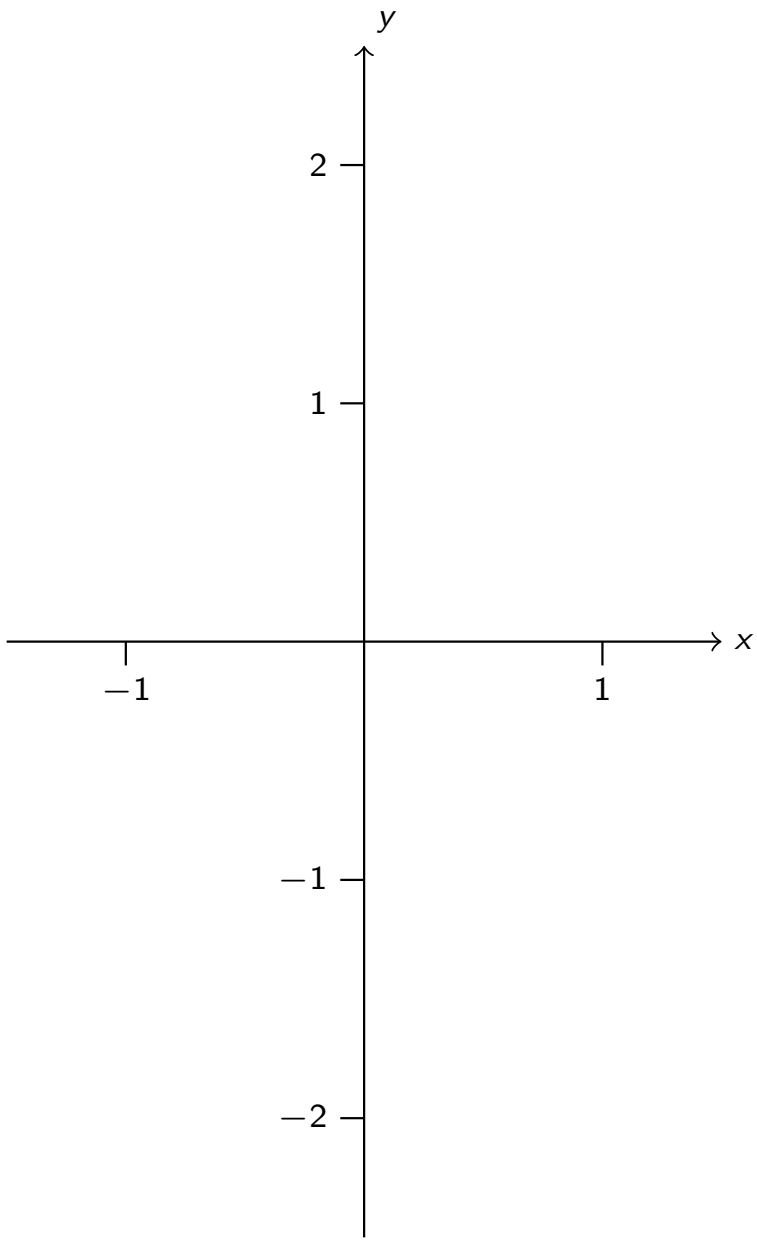
The motion of a particle is described by $\mathbf{r} : \mathbb{R} \longrightarrow \mathbb{R}^2$.

$$\mathbf{r}(t) = \cos(2t)\mathbf{i} - 2\sin(2t)\mathbf{j}, \qquad \text{for } t \geq 0.$$

Sketch:

1. the Cartesian equation of the path;
2. the position of the particle at times
   (i) $t = 0$ (ii) $t = \frac{\pi}{4}$ (iii) $t = \frac{\pi}{2}$;
3. the time taken by the particle to return to its original position;
4. the direction of motion.

4.4 Parametric Curves



4.4 Parametric Curves: Rates of Change